



Turning alphas into betas: Arbitrage and endogenous risk

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ARTICLE INFO

Article history:

Received 6 November 2018

Revised 14 August 2019

Accepted 30 August 2019

Available online 5 March 2020

JEL classification:

G11

G12

G23

Keywords:

Endogenous risk

Factor betas

Financial intermediaries

Arbitrage

Asset pricing anomalies

ABSTRACT

Using data on asset pricing anomalies, I test the idea that the act of arbitrage turns “alphas” into “betas”: Assets with high initial abnormal returns attract more arbitrage and covary endogenously more with systematic factors that arbitrage capital is exposed to. This channel explains the exposures of 40 anomaly portfolios to aggregate funding liquidity shocks and arbitrageur wealth portfolio shocks. My results highlight that financial intermediaries that act as asset market arbitrageurs not only price assets given risks, but also actively shape these risks through their trades.

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¹ I thank John Campbell, Samuel Hanson, Andrei Shleifer, Jeremy Stein, Adi Sunderam, and an anonymous referee for many helpful comments and suggestions. I also thank Tobias Adrian, Vikas Agarwal, Lauren Cohen, Kent Daniel (discussant), Erkki Etula, Wayne Ferson, Will Gornall, Valentin Haddad, Byoung-Hyoun Hwang, Christian Julliard, Yosub Jung, Anastasios Kagkadis (discussant), Paul Karehnke (discussant), Bryan Kelly, Peter Kondor, Dong Lou, Chris Malloy, Ian Martin, David McLean (discussant), Tyler Muir (discussant), David Ng, Christopher Polk, Valery Polkovnichenko (discussant), Emil Siriwardane, Andrea Tamoni, Argyris Tsiaras, Dimitri Vayanos, Michela Verardo, Yao Zeng, and seminar participants at Boston College, Columbia Business School, Cornell University (Dyson), Dartmouth College, Harvard Business School, ITAM, London School of Economics, Rutgers Business School, University of British Columbia, University of Southern California, Hanyang University, Korea University, Seoul National University, Adam Smith Asset Pricing Workshop, AFA (Atlanta), EFA (Warsaw), and University of Washington Summer Finance Conference for comments and discussions. I thank Robert Novy-Marx and Mihail Velikov, who generously shared their data on anomalies. Jonathan Tan, Karamfil Todorov, and Yue Yuan provided superb research assistance. Financial support from the Paul Woolley Centre at the LSE is gratefully acknowledged.

<https://doi.org/10.1016/j.jfineco.2020.02.011>

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1. Introduction

Asset pricing “anomalies” are investment strategies with high expected returns but low identifiable risks. These anomalies—such as value and momentum—first gained widespread recognition among finance academics and investment managers in the early 1990s. Since then, arbitrageurs such as hedge funds have allocated growing amounts of capital to these anomalies.

What is an equilibrium outcome of this arbitrage? Do arbitrageurs eliminate the abnormal returns on these anomalies without affecting their equilibrium risks? Or do anomalies become increasingly exposed to “endogenous risks” because of the very fact that many arbitrageurs are attempting to exploit them?

In this paper, I show empirically that arbitrage activity exposes asset pricing anomalies to endogenous risks associated with the act of arbitrage. The emergence of these endogenous risks means that mispricing in the form of abnormal returns persists in the form of distorted risks.

I draw out testable predictions of this endogenous risk view from intermediary asset pricing models in which fi-

² These models include Gromb and Vayanos (2002, 2018), Brunnermeier and Pedersen (2009), He and Krishnamurthy (2012), He and Krishnamurthy (2013), Brunnermeier and Sannikov (2014), Kozak et al. (2018), and Kondor and Vayanos (2019).

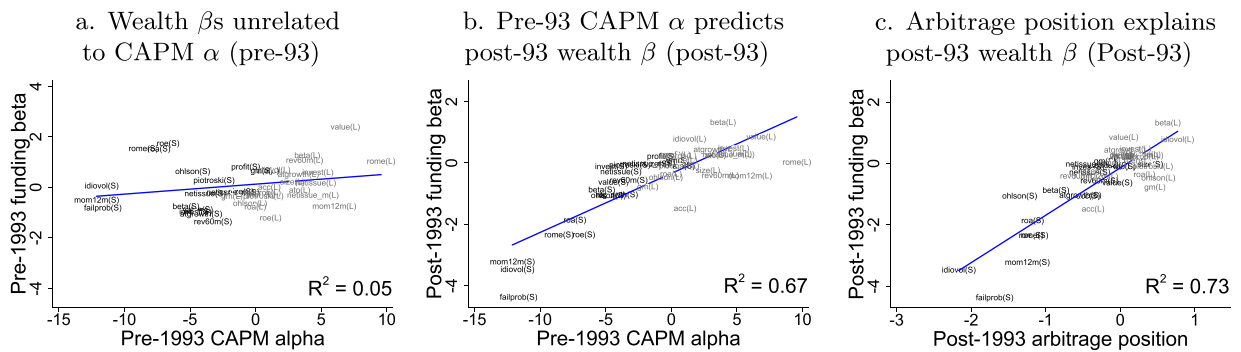


Fig. 2. Explaining the cross-section of arbitrageur wealth portfolio betas.

The first figure shows that the arbitrageur wealth portfolio betas of anomaly portfolios in the pre-1993 period (1974–1993) cluster around zero, irrespective of their CAPM alphas. The next two figures show that the wealth portfolio betas of anomaly portfolios in the post-1993 period (1994–2016) are explained by their pre-arbitrage alphas (pre-1993 CAPM alpha) and post-1993 arbitrage position (inferred from abnormal short positions). Long-side and short-side anomaly portfolios are denoted in gray and black, respectively.

and auxiliary tests also confirm that the wealth portfolio betas of anomaly portfolios are arbitrage-driven.

The strong support for arbitrage-driven risk given by funding liquidity and arbitrageur wealth portfolio betas prompts me to consider whether finding a strong cross-sectional relation between arbitrage variables and factor betas of anomaly portfolios is a low-hurdle test. I therefore calculate the likelihood that betas with respect to a random factor portfolio produce results as strong as mine. I find that only about 1 out of 500 random factors (0.2%) generate results as strong as those of funding liquidity betas and arbitrageur wealth portfolio betas, suggesting that the odds of drawing two factors that by chance both happen to generate my strong results are extremely low.

I conduct a battery of robustness checks. First, using alternative years in the early 1990s as the sample cutoff generates results similar to those I obtain based on the 1993 cutoff. Second, controlling for pre-1993 return volatility or market liquidity does not materially affect my results, implying that my alphas-into-betas result is not a spurious result driven by pre-1993 CAPM alpha proxying for volatility or market liquidity. Third, I use the crash in arbitrage capital during the “quant” crisis of August 2007 to study the determinants of exposure to arbitrage capital shocks without relying on a factor model. I find that both pre-1993 CAPM alpha and post-1993 arbitrage positions strongly explain the cross-section of returns during the crisis, bolstering the evidence that these variables determine equilibrium exposure to arbitrage capital shocks.

My findings suggest that asset pricing anomalies can appear to persist despite arbitrage because the act of arbitrage itself turns their CAPM alphas into intermediary asset pricing betas³. That is, abnormal returns could be low in equilibrium due to arbitrage, but assets could still have high discount rates due to high endogenous arbitrage-driven betas. Hence, abnormal returns could be a poor

proxy for distortions in asset prices (Cohen et al., 2009; Cho and Polk, 2020).

My finding that the act of arbitrage turns alphas into betas and has transformed the variance-covariance structure of asset pricing anomalies relates to findings in other studies. McLean and Pontiff (2016) and Dong et al. (2018) find that correlations among anomaly portfolios change around academic publication, which the authors attribute to arbitrage trades. My results imply that changes in systematic funding liquidity and arbitrageur wealth portfolio exposure are important drivers of this correlation change. Drechsler and Drechsler (2016) find that anomaly arbitrageurs face risk concentrated in negative alpha stocks and require a return premium for bearing this risk. I find that the role of arbitrageurs extends beyond recognizing such risk; they also propagate it to other assets they trade. Other evidence that institutional trades affect the second and third moments of asset returns includes Barberis et al. (2005), Brunnermeier et al. (2008), Greenwood and Thesmar (2011), Anton and Polk (2014), Ben-David et al. (2018) and Lou and Polk (2019). Relatedly, Liao (2019) documents another unintended consequence of arbitrage: the act of arbitrage on one anomaly (such as credit spread differential) can affect an anomaly in a different market (such as covered interest parity deviation).

My results suggest that financial intermediaries themselves determine the cross-section of asset betas with respect to intermediary asset pricing factors. Hence, intermediary-based asset pricing is a framework that gives a complete account of not only which risks are priced in the cross-section (Adrian et al., 2014; He et al., 2017; Avdjiev et al., 2019), but also how these risks arise in the first place.

Finally, my findings are broadly consistent with the core message of the “adaptive market hypothesis” (Lo, 2004) that the expected returns and risks of trading strategies change over time as a new class of arbitrageurs, such as

³ McLean and Pontiff (2016) show a 32% decline in the returns of 97 anomalies after their publication. Chordia et al. (2014) also find that anomaly returns have not completely disappeared.

hedge funds, interacts with existing investors.⁴ Evidence for this hypothesis has heretofore been qualitative; my analysis provides quantitative evidence.

2. Methodology and data

I explain the theoretical motivation for my empirical tests and the data I use to conduct the tests.

2.1. Theoretical background

I provide a simple theoretical framework from which arise the predictions I test in the data. The framework draws heavily on the continuous time model of Kondor and Vayanos (2019), but interested readers can find the same predictions in a three-period setting in the Online Appendix to this paper.⁵

Consider a continuous time economy in which uncertainty is captured by the N -dimensional Brownian motion B_t and the risk-free rate is fixed at $r > 0$. There are N risky assets in zero net supply with cash flows

$$dD_t = \bar{D}dt + dB_t, \quad (1)$$

where $\bar{D}$ is a constant $N \times 1$ vector. Hence, asset $i \in \{1, \dots, N\}$ has a unit fundamental cash flow exposure to the Brownian motion i .⁶ There are two types of investors, households and arbitrageurs. Households receive a random endowment of $u^T dD_t$ at $t + dt$ for some $N \times 1$ constant vector u and are mean variance optimizers of instantaneous changes in wealth at $t + dt$, where $A > 0$ is their coefficient of absolute risk aversion.

Arbitrageurs do not face endowment shocks, so they perceive u as distortions in asset demands that generate abnormal return opportunities. Arbitrageurs maximize power utility over consumption,

$$E_0 \left[\int_0^\infty e^{-\rho t} \frac{c_t^{1-\gamma}}{1-\gamma} dt \right], \quad (2)$$

where c_t is instantaneous consumption, $\rho > \gamma$ is the discount rate, and $\gamma \geq 0$ is the coefficient of relative risk aversion.⁷ The capital of arbitrageurs k_t evolves according to

$$dk_t = rk_t dt + x_t^T dR_t - c_t dt, \quad (3)$$

where x_t is the arbitrageur's position in risky assets and dR_t denotes risky asset returns in excess of the risk-free rate.⁸

What does this setup imply about asset returns when the market is occupied solely by households? This proxies for a sample period in which arbitrageurs are small or, for

whatever reason, have not engaged in arbitrage trades on assets.

Proposition 1. In the absence of arbitrageurs, expected excess returns on assets follow

$$\frac{E_t[dR_t]}{dt} = Au. \quad (4)$$

Proof. See Kondor and Vayanos (2019) for the proofs of this and the next propositions. $\square$

Intuitively, without arbitrageurs, assets with greater exposure to endowment shocks generate higher expected returns and equilibrium risk is determined solely by fundamental cash flows.

On the other hand, when arbitrageurs enter the market, the act of arbitrage leads asset returns to covary endogenously with arbitrage capital.

Proposition 2. In the presence of arbitrageurs, arbitrage positions x_t , endogenous return covariances with arbitrage capital, and expected excess returns of assets are given by

$$x_t = \tilde{x}(k_t)u, \quad (5)$$

$$\frac{\text{Cov}_t(dR_t, dk_t)}{dt} \propto u, \quad (6)$$

$$\frac{E_t[dR_t]}{dt} = \mu(k_t)u, \quad (7)$$

where $\tilde{x}(k_t) > 0$ and $\mu(k_t) > 0$ are increasing and decreasing in k_t , respectively.

Intuitively, arbitrageurs in equilibrium play a larger price-correcting role in assets with larger demand distortions and greater abnormal returns (Eq. (5)). This, however, also means that such assets are endogenously more sensitive to variation in the price-correcting role coming from arbitrage capital shocks (Eq. (6)). Arbitrageurs take this endogenous covariance into account when they determine the expected returns on assets (Eq. (7)).

Hence, an asset with a larger demand distortion u attracts a greater arbitrage position and develops a correspondingly larger endogenous covariance with arbitrage capital. Although u may not be observable, it is revealed by abnormal returns in the absence of arbitrageurs (Au in Eq. (4)) or by equilibrium arbitrage position in the presence of arbitrageurs ($\tilde{x}(k_t)u$ in Eq. (5)). Hence, holding all else fixed, an abnormal return in the absence of arbitrageurs predicts endogenous risk exposure to arbitrage capital shocks (i.e., alphas turn into betas). When approximated as a linear relation, this implies a cross-sectional regression:

$$\beta_{i,k} = b_0 + b_1 \alpha_i^{\text{pre}} + u_i, \quad (8)$$

where $\beta_{i,k} \propto \frac{\text{Cov}_t(dR_t, dk_t)}{dt}$ is the beta exposure to k_t at $t + dt$ and $\alpha_i^{\text{pre}} \equiv Au$ is “pre-arbitrage” abnormal return. Alternatively, equilibrium arbitrage position explains the contemporaneous endogenous risk exposure to k_t , motivating a regression:

$$\beta_{i,k} = \tilde{b}_0 + \tilde{b}_1 x_i + \tilde{u}_i, \quad (9)$$

⁴ The expression “alphas becoming betas” was first used to describe this hypothesis.

⁵ Available at <https://sites.google.com/site/thummimcho>.

⁶ All results hold analogously when $dD_t = \bar{D}dt + \sigma^T dB_t$ with $\sigma^T \neq I_N$, but the independence assumption simplifies notation.

⁷ $\rho > \gamma$ prevents arbitrageurs from accumulating infinite wealth over time.

⁸ Although shocks to k_t come from shocks to the wealth portfolio of arbitrageurs, they can be interpreted more broadly as coming from all systematic arbitrage capital shocks, including aggregate funding liquidity shocks.

Table 1

List of 40 anomaly portfolios.

This table describes the 40 anomaly portfolios used in the paper. α_{CAPM}^{pre} is CAPM alpha over 1974m1–1993m12. Boldface denotes estimated alphas greater than 1.96 times the heteroskedasticity robust standard error in absolute value. Mktcap share is the portfolio's total market capitalization over total market capitalization of all domestic common US stocks listed on the NYSE, Amex, and Nasdaq, averaged over 1974m1–2016m12.

Type	Long (top decile)				Short (bottom decile)				Academic publication	
	No	Label	α_{CAPM}^{pre}	Mktcap share	No	Label	α_{CAPM}^{pre}	Mktcap share	Year	Sample
Beta arbitrage	1	beta(L)	3.9	0.09	21	beta(S)	−5.3	0.09	1973	1926–1968
Return on market equity	2	rome(L)	9.6	0.05	22	rome(S)	−8.6	0.03	1977	1956–1971
Ohlson's O-score	3	ohlson(L)	−0.4	0.29	23	ohlson(S)	−4.8	0.01	1980	1970–1976
Size	4	size(L)	2.8	0.02	24	size(S)	−1.1	0.58	1981	1926–1975
Long-run reversals	5	rev60m(L)	3.7	0.03	25	rev60m(S)	−3.3	0.13	1985	1926–1982
Value	6	value(L)	6.8	0.04	26	value(S)	−4.4	0.20	1985	1980–1990
Momentum	7	mom12m(L)	6.0	0.10	27	mom12m(S)	−12.1	0.04	1990	1964–1987
Net issuance	8	netissue(L)	4.6	0.11	28	netissue(S)	−3.8	0.08	1995	1980–1990
Net issuance monthly	9	netissue_m(L)	4.4	0.11	29	netissue_m(S)	−1.7	0.09	1995	1980–1990
Accruals	10	acc(L)	1.0	0.06	30	acc(S)	−4.6	0.05	1996	1962–1991
Return on assets	11	roa(L)	−0.0	0.17	31	roa(S)	−7.4	0.03	1996	1979–1993
Return on book equity	12	roe(L)	1.1	0.14	32	roe(S)	−6.7	0.04	1996	1979–1993
Failure probability	13	failprob(L)	0.5	0.16	33	failprob(S)	−11.6	0.02	1998	1981–1996
Piotroski's f-score	14	piotroski(L)	0.6	0.21	34	piotroski(S)	−3.2	0.09	2000	1976–1997
Investment	15	invest(L)	4.7	0.03	35	invest(S)	−4.6	0.07	2004	1973–1996
Idiosyncratic volatility	16	idiov(L)	1.4	0.25	36	idiov(S)	−11.7	0.04	2006	1986–2000
Asset growth	17	atgrowth(L)	3.3	0.03	37	atgrowth(S)	−4.2	0.10	2008	1968–2003
Asset turnover	18	ato(L)	3.4	0.05	38	ato(S)	0.9	0.09	2008	1984–2002
Gross margins	19	gm(L)	−1.8	0.20	39	gm(S)	0.5	0.04	2008	1984–2002
Gross profitability	20	profit(L)	0.4	0.10	40	profit(S)	−0.8	0.07	2010	1976–2005

where x_i is the average arbitrage position on asset i over the period in which the beta is measured. These are the main cross-sectional restrictions I test in the data.

Endogenous covariance with arbitrage capital displays three additional patterns I test in the data. First, the endogenous covariance is hump-shaped with respect to arbitrage capital k_t . It is small when k_t is close to zero since arbitrageurs only have a small impact on prices. But it is also small when k_t is close to infinity, in which case the assets are priced almost exclusively by arbitrageurs, whose absolute risk aversion remains close to zero for small changes in k_t .⁹ Hence, endogenous covariance with arbitrage capital tends to arise when arbitrageurs hold non negligible capital but are constrained and disappear when arbitrageurs are unconstrained. Second, in the context of the return decomposition of Campbell and Shiller (1988) and Campbell (1991), this covariance arises from the expected returns (discount rates) of assets covarying with k_t . Hence, holding all else constant, assets with a larger endogenous covariance with arbitrage capital have more volatile expected returns and feature greater time-series predictability in returns. Third, endogenous covariance with arbitrage capital can be observed in a crash in arbitrage capital. In a large enough crash, the magnitude of the return response is greater in assets with a larger demand distortion u since the arbitrageur plays a larger price-correcting role in the asset.

2.2. Data and measurement

My test assets are 40 anomaly portfolios formed by taking the long and short portfolios (top/bottom deciles) from

a univariate sorting of stocks on 20 anomaly characteristics (see the list in Table 1). These 20 characteristics, constructed based on data from the Center for Research in Security Prices and Compustat, represent a standard set of low-turnover anomaly characteristics.¹⁰ One can arrive at this set by taking the 32 characteristics surveyed by Novy-Marx and Velikov (2016) and excluding the 5 redundant (e.g., “high-frequency combo”) and 7 highest turnover (e.g., short-term reversal) characteristics. I exclude high-turnover anomalies since arbitrage-driven betas should not arise in anomalies with a short mispricing horizon (Kondor and Vayanos, 2019). I use long and short portfolios as separate test assets, rather than forming long-short portfolios, for two reasons: (a) actual arbitrageurs typically do not form a long-short portfolio based on single anomaly characteristic but consider multiple anomaly characteristics of stocks and (b) it ensures a large cross-sectional variation in the right-hand variable (e.g., arbitrage position and pre-arbitrage alpha), which increases the power of the test. I determine characteristic deciles based on NYSE stocks. I compute monthly rebalanced value-weighted portfolio returns based on domestic common stocks listed on the three major exchanges (NYSE, Amex, and Nasdaq). For analyses requiring quarterly portfolios, I use rolled over one-month returns to obtain quarterly returns. My sample period is from 1974 to 2016.¹¹

I study two kinds of systematic shocks to arbitrage capital. The first is the aggregate funding liquidity factor of Adrian et al. (2014), calculated as shocks to the book leverage of security broker-dealers as a factor cap-

¹⁰ The Online Appendix to this paper provides more information on the construction of anomaly characteristics.

¹¹ Like Novy-Marx and Velikov (2016), I do not use data from before the early 1970s because of the poor quality of quarterly accounting data.

⁹ This prediction also arises in (Brunnermeier and Pedersen, 2009).

turing aggregate funding liquidity shocks.¹² Since anomaly arbitrageurs such as quantitative long/short equity hedge funds rely on leverage and hence funding liquidity, they are likely to be exposed to the aggregate funding liquidity factor (Brunnermeier and Pedersen, 2009; Aragon and Strahan, 2012 and Mitchell and Pulvino, 2012). This seems especially plausible since security broker-dealers provide funding to hedge funds as their prime brokers. The second is shocks to the aggregate arbitrage portfolio proxied by a value-weighted portfolio that goes long on the top decile and short on the bottom decile of stocks sorted on my measure of arbitrage position explained below.¹³ To prevent microcaps from dominating the decile portfolios, the decile cutoffs are determined by the distribution of arbitrage positions on NYSE stocks. I estimate exposure to these shocks as the beta with respect to the shocks in a portfolio-specific time-series regression that controls for market exposure. The funding liquidity factor and arbitrageur wealth portfolio returns have a small positive correlation of 0.075 over the sample period, implying that they capture different components of arbitrage capital shocks.

I measure arbitrage positions on portfolios using abnormal short positions on underlying stocks, following (Ben-David et al., 2012; Boehmer et al., 2013; Hanson and Sunderam, 2014; Hwang et al., 2018), among others.¹⁴ Since most short positions are held by hedge funds (approximately 85%, according to a Goldman Sachs report), an abnormally high (low) short position on an anomaly signals a net short (long) position taken by the arbitrageurs.¹⁵ Each month, I estimate abnormal short position on each stock as the residual from a cross-sectional regression of short interest ratio (shares shorted divided by shares outstanding) from Compustat on ten dummy variables for size deciles, ten dummy variables for liquidity deciles (determined by Amihud (2002) measure over the prior 12 months), and two dummy variables for convertible bonds outstanding (one for having any convertible bonds outstanding and the other for it being greater than 10% of

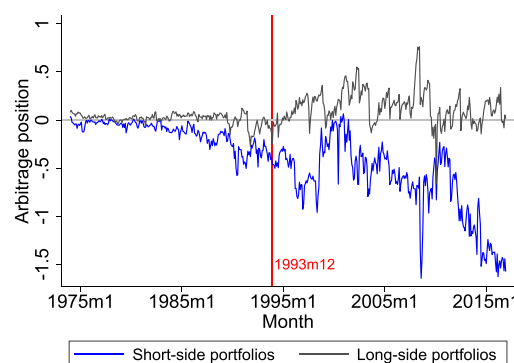


Fig. 3. Estimated arbitrage positions.

The figure plots the equal-weighted (cross-sectional) average of arbitrage positions in long-side and short-side anomaly portfolios over 1974m1–2016m12. Arbitrage position is inferred from abnormal short positions (see Section 2.2).

the market capitalization of common equity).¹⁶ I control for convertible bonds outstanding to net out the effect of convertible bond arbitrage. I also remove stocks involved in mergers and acquisitions, as defined in (Jiang et al., 2018), to net out the effect of merger arbitrage (risk arbitrage). Then, I compute arbitrage position on a portfolio as the negative (-1×100 to express it as a percentage) of the value-weighted average of abnormal short positions on underlying stocks.

Following Schwert (2003), I use the post-1993 period (1994–2016) as the period in which the anomaly portfolios are actively traded by arbitrageurs such as the quantitative equity hedge funds. The growth of arbitrage on anomalies around 1993, also observed in (Chordia et al., 2008; 2011; Stein, 2009; Chordia et al., 2014), is likely due to a combination of factors including the growth in the hedge fund industry, publication of seminal works on cross-sectional anomalies (e.g., Fama and French, 1993; Jegadeesh and Titman, 1993), and improved market liquidity. I do not intend to differentiate among these channels but interpret my post-1993 results as driven by a combination of these factors. Fig. 3 shows that arbitrage positions on anomaly portfolios indeed did grow rapidly around 1993. Although not reported in the paper, I also find that average position on anomaly portfolios tends to fall in times of negative funding liquidity shocks in the post-1993 period but not in the pre-1993 period.

The theoretical framework in Section 2.1 shows that besides arbitrage position, abnormal return in the absence of arbitrageurs is another predictor of endogenous exposure to arbitrage capital shocks. Building on the evidence that hedge fund flow chases CAPM alpha (Agarwal et al., 2018), I use CAPM alpha in the pre-1993 period (1974–1993) to proxy for abnormal return in the absence of arbitrageurs.¹⁷

¹² Specifically, the factor is the change in the log of aggregate leverage of the entire broker-dealer sector obtained from the Federal Reserve Board's flow of funds data. The book leverage is adjusted for seasonality before taking the growth rate. Adrian et al. refer to the factor as a "leverage" factor, but I refer to it as a funding liquidity factor since it is meant to capture aggregate funding liquidity shocks à la Brunnermeier and Pedersen (2009). (A related paper by Asl and Etula, 2012 calls it a funding factor.) I extend the quarterly funding factor to 2016q4 and standardize it over my sample period 1974q1–2016q4. See Adrian et al. (2014) and the Online Appendix to this paper for detailed instructions on constructing the series.

¹³ I thank the anonymous referee for the idea of using my arbitrage position measure to reconstruct the arbitrageur wealth portfolio.

¹⁴ Although Chen et al. (2018) show that equity positions of hedge funds can also be inferred from institutional holdings (13F) data, it is difficult to infer positions of quantitative long/short equity hedge funds that trade anomalies since 13F filings are available only at the holding company level. Using the data on long hedge fund positions kindly provided by Chen et al. (2018), I find that aggregate hedge fund long positions inferred from the 13F do not have a statistically significant relation to past alphas, implying that these are unlikely to represent the long positions of anomaly arbitrageurs.

¹⁵ "Goldman Sachs hedge fund trend monitor" (February 20, 2008)

¹⁶ I use short interests reported in mid-month and shares outstanding on the same day (if available) or the previous trading day. The exact method used to obtain abnormal short positions, however, does not affect my results.

¹⁷ Using a two-factor alpha that removes the possible risk premium associated with the arbitrage-related factor (funding liquidity or arbitrage portfolio) or alternative multifactor alphas generates similar results. For

Table B.1 shows that pre-1993 CAPM alpha strongly predicts arbitrage activity in the post-1993 period (see the coefficient on $\alpha^{pre} \times \text{Post-1993}$), suggesting that demand distortion in an anomaly portfolio revealed by pre-1993 CAPM determines the amount of post-1993 arbitrage activity on the portfolio.¹⁸ The table also highlights the potential role of academic publication, which I include as a regressor in my panel analysis.

Besides the act of arbitrage, fundamental cash flows may also contribute to equilibrium risk exposure.¹⁹ To account for cash flow exposure to risk, I use pre-1993 factor beta or fundamental characteristics of the anomaly portfolio (size, book-to-market ratio, profitability, and investment characteristics) as additional controls. A portfolio's characteristic is defined as the value-weighted average characteristic decile of the underlying stocks, where the deciles are determined by the distribution of NYSE stocks.²⁰

2.3. Standard errors

Throughout the paper, I compute standard errors based on a bootstrap procedure that takes into account both cross-portfolio covariances and generated regressors. I use bootstrapping instead of generalized method of moments (GMM) since to my knowledge, existing GMM approaches cannot be used in my case, in which different variables are estimated using data with different frequencies. However, for the main regressions in Table 6 (Columns (1) and (4)), which only use data with monthly frequency, I find that GMM standard errors are smaller than my bootstrap standard errors, implying that my bootstrap t -statistics are conservative. The t -statistics based on GMM are 3.05 and 3.60 for Columns (1) and (4) of Table 6, and those based on my bootstrap standard errors are 2.73 and 3.13, respectively.²¹ My analysis on randomly generated factors in Table 8 also suggests that my bootstrap t -statistics are conservative; the one-sided p -values implied by my t -statistics in Columns (1) and (4) of Table 6 (0.32% and 0.09%) are higher than those implied by the random factors (0.20% and 0.05%). See Appendix A for further details on my bootstrap procedure.

failure probability anomaly, I use CAPM alpha over 1981–1993 to account for the anomaly's sensitivity to sample period, emphasized in (Dichev, 1998).

¹⁸ My finding on the 1993 cutoff is somewhat at odds with the finding that no return decay is observed in the anomalies following 1993 (McLean and Pontiff, 2016). The main reason for this difference is that short interest measures the arbitrage activity of a group of sophisticated arbitrageurs, whereas return decay reflects investment by all types of investors. Another contributing factor is that I use the year in which the anomaly was first published, not when it was first well publicized. For example, the academic publication of the value anomaly is (Rosenberg et al., 1985) in my data, but it is (Fama and French, 1992) in McLean and Pontiff.

¹⁹ In the framework presented in Section 2.1, this is reflected in the equilibrium asset return, $dR_t = \mu(k_t)u dt + (g(k_t)u + 1)dB_t$ for some stochastic terms $\mu(k_t)$ and $g(k_t)$, which implies that asset return has an endogenous exposure to risk $g(k_t)u dB_t$ in addition to fundamental cash flow exposure dB_t . See Kondor and Vayanos (2019) for further details.

²⁰ See the Online Appendix for more information about how I construct these characteristics for each stock.

²¹ The MATLAB code for GMM standard errors is available upon request.

3. Funding liquidity exposure

Anomaly portfolios display a large cross-sectional variation in funding liquidity exposure in the post-1993 period (Fig. 1). In this section, I provide evidence that this variation arises endogenously through the act of arbitrage.

3.1. Cross-section of funding betas

The starting evidence is a strong contemporaneous cross-sectional relation between arbitrage position and funding liquidity beta (Eq. (9) in Section 2.1). Columns (1)–(3) of Table 2 show that anomaly portfolios with greater arbitrage positions tend to have higher funding betas in the post-1993 period. In terms of magnitude, a 1 percentage point (pp) rise in the fraction of shares outstanding held by arbitrageurs (which is what the arbitrage position captures) increases the portfolio's funding beta by 1.6–1.8, which corresponds to portfolio return responding 1.6–1.8 pp more to a one standard deviation shock in aggregate funding liquidity.²² This is consistent with the prediction that portfolios in which arbitrage capital plays a larger price-correcting role respond more to the variation in arbitrage capital due to funding liquidity shocks. Arbitrage position alone explains more than 70% of the cross-sectional variation in post-1993 funding betas, and neither pre-1993 funding beta nor fundamental characteristics contribute to the cross-sectional fit. This suggests that the cross-sectional variation in funding liquidity exposure is mostly arbitrage driven.

However, using arbitrage position as the right-hand variable raises a reverse causality concern: arbitrageurs may take larger positions on stocks with larger funding betas to earn an extra risk premium associated with funding liquidity (Jurek and Stafford, 2015).²³ A remedy is to use pre-1993 CAPM alpha as a right-hand variable since it measures the demand distortion in the anomaly portfolio that ultimately determines the equilibrium arbitrage position and since an alpha—when correctly measured—captures the part of expected return unrelated to factor exposure.²⁴ Estimating the relation in Eq. (8) in Section 2.1, Columns (4)–(6) show that pre-arbitrage alpha strongly explains the cross-section of funding betas. A 1 pp increase in pre-1993 CAPM alpha raises post-1993 funding liquidity beta by 0.19–0.21, and pre-1993 CAPM alpha alone explains 67% of the cross-sectional variation post-1993 funding liquidity exposure of anomaly portfolios. This, together with the result based on the arbitrage position, implies that (pre-1993 CAPM) alpha turned into (post-1993 funding liquidity) beta through the act of arbitrage.

In the pre-1993 period, arbitrage position does not explain the cross-section of funding betas, consistent with

²² The funding liquidity factor I use has been standardized to have a standard deviation of one.

²³ Relatedly, Song (2017) finds that mutual funds with lower skills increase factor exposure.

²⁴ One could argue that pre-1993 CAPM alpha includes a risk premium for funding liquidity exposure, in which case pre-1993 CAPM alpha would predict post-1993 funding beta almost mechanically. However, using the pre-1993 two-factor alpha that controls for the possible risk premium associated with funding liquidity generates very similar results.

Table 2

Explaining the cross-section of funding liquidity betas.

Baseline: $\beta_{funding,i}^{post93} = b_0 + b_1 \text{Arbitrage position}_i^{post93} + u_i$

This table shows that the post-1993 funding liquidity betas of 40 anomaly portfolios can be cross-sectionally explained by arbitrage position or pre-1993 CAPM alpha, whereas pre-1993 funding liquidity betas do not display such patterns. Unless otherwise noted, the right-hand variables are calculated for the same sample period as the left-hand variable. Funding liquidity betas are betas with respect to the funding liquidity factor of [Adrian et al. \(2014\)](#), estimated in a two-factor model that includes the market factor. Arbitrage position is inferred from abnormal short position on underlying stocks. Characteristic ranks are value-weighted decile ranks of the underlying stocks' characteristics. Pre-1993 and post-1993 periods are 1974–1993 and 1994–2016, respectively. In the parentheses are *t*-statistics based on bootstrap standard errors that account for cross-portfolio covariances as well as generated regressors. Boldface denotes coefficient estimates with the absolute value of *t*-statistics greater than 1.96.

	$\beta_{funding}^{post93}$								$\beta_{funding}^{pre93}$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Arbitrage position	1.55 (2.31)	1.57 (2.33)	1.80 (2.06)						0.68 (0.28)		
α_{CAPM}^{pre93}				0.19 (2.62)	0.20 (2.54)	0.21 (2.27)				−0.03 (−0.48)	
$\beta_{funding}^{pre93}$		0.24 (1.07)			−0.09 (−0.28)		0.16 (0.39)				
Size rank			0.10 (0.35)			−0.14 (−0.52)		−0.41 (−1.17)	−0.03 (−0.11)	0.05 (0.29)	0.02 (0.10)
Value rank			0.11 (0.48)			−0.16 (−0.48)		0.37 (1.50)	0.45 (1.57)	0.54 (1.71)	0.48 (1.82)
Profitability rank			−0.04 (−0.24)			−0.14 (−0.65)		0.19 (0.83)	0.07 (0.37)	0.13 (0.60)	0.09 (0.46)
Investment rank			−0.06 (−0.36)			−0.10 (−0.55)		0.16 (0.99)	−0.07 (−0.51)	−0.11 (−0.93)	−0.09 (−0.75)
Constant	−0.14 (−0.74)	−0.16 (−0.80)	−0.31 (−0.16)	−0.34 (−1.13)	−0.33 (−1.13)	2.04 (0.69)	−0.56 (−1.56)	−2.71 (−0.99)	−2.71 (−1.13)	−3.48 (−1.27)	−2.89 (−1.30)
Observations	40	40	40	40	40	40	40	40	40	40	40
R_{adj}^2	0.72	0.75	0.75	0.67	0.66	0.71	−0.01	0.28	0.71	0.72	0.71

my interpretation of the pre-1993 period as the period when anomaly arbitrageurs were small (Columns (9)–(11)). Instead, fundamental characteristics of anomaly portfolios explain around 70% of the cross-sectional variation in their funding liquidity exposure. The large magnitude associated with value rank suggests that portfolios with a large value characteristic (high book-to-market ratio) tend to have large funding liquidity exposure in the pre-1993 period, possibly because these are distressed stocks whose cash flows depend importantly on the availability of short-term funding in the financial sector. However, the effect of each individual characteristic is not statistically significant.

3.2. Panel of funding betas

An alternative to the purely cross-sectional approach is to study a panel (portfolios $\times$ time) of funding betas. To do this, I use time-varying betas estimated from a moving window of ± 14 quarters (7 years) around each quarter as the left-hand variable. As a right-hand variable, I use arbitrage position, pre-1993 CAPM alpha interacted with a post-1993 dummy, and pre-1993 CAPM alpha interacted with a post publication dummy. As additional controls, I include (a) anomaly fixed effects to control for unobserved heterogeneity across the portfolios that may be correlated with funding betas, (b) all variables used to create interaction terms (except those subsumed by anomaly fixed effect), (c) time-varying fundamental characteristics to control for changes in the characteristics that may affect the betas, and (d) quadratic time trends to control for average trends in the betas.

Table 3 shows results consistent with the cross-sectional result. A time-series increase in the arbitrage position leads to an increase in the funding beta (Columns (1)–(2)). The coefficient is around half of that implied by the cross-sectional regressions, which suggest that not taking the time-series average leads to attenuation due to the measurement error in the estimated arbitrage position. Funding beta more specifically changes around post-1993 with the direction and magnitude implied by the pre-1993 CAPM alpha, and the estimated effect of 0.18 is close to 0.19 estimated in the cross-section (Columns (3)–(4)). Hence, the panel approach also suggests that alphas turned into betas through the act of arbitrage.

Albeit with slightly lower *t*-statistics, pre-1993 CAPM alpha also affects funding betas around the academic publication of the anomalies (Columns (5)–(6)). However, it is difficult to disentangle the effect of pre-1993 CAPM alpha around 1993 into that coming from the increase in arbitrage capacity due to the growth of hedge fund capital in the 1990s and improved market liquidity and that coming from the identification new anomalies since the early 1990s (Column (7)). This is consistent with my interpretation of the post-1993 effect as coming from a combination of those events. The two-stage least squares regressions in Columns (9)–(11) study the effect of arbitrage position changes driven by pre-1993 CAPM alphas around 1993 and by academic publication. The large estimated coefficient suggests that the time series of arbitrage position is indeed a noisy measure of actual arbitrage position and the two-stage approach identifies the effect of estimated arbitrage position changes that are more purely driven by actual arbitrage.

Table 3

Explaining the panel of funding liquidity betas.

$$\text{Baseline: } \beta_{\text{funding},i,t} = b_0 + b_1 \text{Arbitrage position}_{i,t} + b'_2 X_{i,t} + b_4 t + b_5 t^2 + u_i + \epsilon_{i,t}$$

$$(X \equiv \text{time-varying characteristics})$$

This table uses a panel regression to show that arbitrage position and pre-arbitrage alpha explain the panel of funding liquidity betas of anomaly portfolios (40 portfolios $\times$ 1974q1–2016q4). For each anomaly portfolio, quarterly funding liquidity betas are estimated in a moving window of ± 14 quarters (7 years) surrounding each quarter. Pre-1993 CAPM alpha interacted with post-1993 and post publication dummies are used as proxies or instruments for arbitrage position. Arbitrage position is inferred from abnormal short position on underlying stocks. A post-1993 dummy, a post publication dummy, quadratic time trends (t and t^2), and a constant are included in the regression (whenever appropriate) but are not reported in the table. In the parentheses are t -statistics based on bootstrap standard errors that account for cross-portfolio covariances, generated regressors, and serial correlations. Boldface denotes coefficient estimates with the absolute value of t -statistics greater than 1.96.

	OLS								2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Arbitrage position	0.70 (2.09)	0.71 (2.11)							3.16 (1.99)	3.08 (1.83)	3.40 (1.84)
$\alpha_{\text{CAPM}}^{\text{pre93}} \times \text{Post-1993}$			0.18 (2.02)	0.18 (2.10)			0.11 (1.51)				
$\alpha_{\text{CAPM}}^{\text{pre93}} \times \text{Post publication}$					0.21 (1.92)	0.22 (2.08)	0.13 (1.46)				
Size rank		−0.03 (−0.20)		−0.12 (−0.79)		−0.18 (−1.07)	−0.15 (−0.98)	−0.12 (−0.73)	0.27 (0.87)	0.26 (0.79)	0.30 (0.87)
Value rank		0.17 (0.88)		0.18 (0.99)		0.15 (0.82)	0.20 (1.07)	0.14 (0.68)	0.28 (1.54)	0.28 (1.49)	0.29 (1.59)
Profitability rank		0.22 (1.32)		0.20 (1.23)		0.21 (1.32)	0.21 (1.33)	0.21 (1.27)	0.23 (1.29)	0.23 (1.26)	0.23 (1.29)
Investment rank		0.22 (1.52)		0.22 (1.35)		0.27 (1.81)	0.24 (1.49)	0.23 (1.54)	0.20 (1.05)	0.20 (1.03)	0.19 (1.00)
Anomaly FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,760	5,760	5,760	5,760	5,760	5,760	5,760	5,760	5,760	5,760	5,760
R_{adj}^2	0.13	0.16	0.20	0.23	0.18	0.22	0.26	0.09	.	.	.
<i>Instrumental variables</i>											
$\alpha^{\text{pre}} \times \text{Post-1993}$									✓	✓	
$\alpha^{\text{pre}} \times \text{Postpub}$									✓		✓

It is interesting to relate my panel regression result to the finding that academic publication increases the anomaly's correlation with other published anomalies (McLean and Pontiff, 2016; Dong et al., 2018). My result shows that the increased correlation arises partly from an increased exposure to the funding factor and that the post publication increase in the beta (correlation) has a cross-sectional pattern predicted by intermediary-based asset pricing models; the increase is larger for an anomaly with a larger pre-arbitrage alpha.

3.3. Funding betas during constrained versus unconstrained periods

What is an alternative explanation for my findings on funding liquidity betas? Suppose that the funding liquidity factor actually proxies for the arbitrageur wealth portfolio rather than aggregate funding liquidity shocks. In this case, even if arbitrageurs were too small to affect the covariances of anomaly returns, a portfolio with a larger arbitrage position or pre-arbitrage alpha may mechanically have a higher beta with the factor since the portfolio would likely be a larger part of the arbitrageur wealth portfolio.

A prediction that can help refute this alternative explanation is that arbitrage-driven betas arise primarily when arbitrageurs are constrained. If funding betas were arbitrage-driven betas, they would arise primarily when arbitrageurs are constrained such that shocks to their capital that relax or tighten their constraint generate variation

in arbitrage positions in the anomaly portfolios. In contrast, if funding betas were mechanical wealth portfolio betas, they would strengthen when arbitrageurs are unconstrained and can hold more anomalies in their portfolios.

To test this prediction, I define constrained and unconstrained periods in two ways. First, I follow Nagel (2012) to proxy constrained (unconstrained) times for institutional arbitrageurs as quarters in which the moving average of the VIX is above (below) the sample median.²⁵ Second, since abnormal returns are likely to be competed away in times when arbitrageurs are unconstrained, I use years in which the anomaly portfolios' alphas re emerge (disappear) as the constrained (unconstrained) times. Specifically, I use years in which CAPM alphas estimated from daily data have a cross-sectional R^2 with pre-1993 CAPM alphas above the median.²⁶ Fig. 4 plots the constrained and unconstrained post-1993 quarters (or years) defined by the two methods. Despite some differences, they both identify the dot-com crash of 2000–2002 and the financial crisis of 2008–2009 as the periods in which arbitrageurs were constrained.

²⁵ I use the exponential-weighted moving average with a smoothing factor of 0.3. However, since the quarterly Chicago Board Options Exchange Volatility Index ("VIX") tends to be persistent, using the original quarterly VIX series delivers similar results.

²⁶ Theoretically, the correct alpha to use here should additionally control for the risk premium associated with arbitrage-driven beta. Yearly alphas that additionally account for exposure to the mimicking portfolio of the funding factor does not change leads to similar classification of constrained times, so I prefer using yearly CAPM alphas for simplicity.

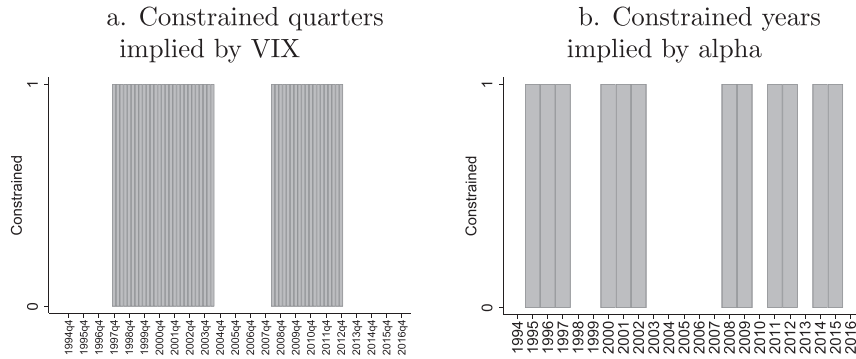


Fig. 4. Proxies for constrained versus unconstrained periods.

The first figure reports constrained (unconstrained) post-1993 quarters defined as quarters in which an exponentially weighted moving average of the VIX (smoothing factor: 0.3) is above (below) the sample median. The second figure reports constrained (unconstrained) years defined as years in which the CAPM alphas estimated from daily data are cross-sectionally explained by pre-1993 alphas with a high R^2 (above median among all year-specific cross-sectional R^2 s).

Table 4

Funding liquidity betas arise during constrained times.

Baseline: $\beta_{funding,i}^{post93,constrained} = b_0 + b_1 \text{Arbitrage position}_i^{post93} + b_2 \beta_{funding,i}^{pre93} + u_i$

This table shows that post-1993 funding liquidity betas of anomalies strengthen in periods in which arbitrageurs are likely to be constrained and weaken when they are likely to be unconstrained, consistent with the predictions of intermediary-based asset pricing models. I define constrained (unconstrained) times for institutional arbitrageurs as (a) quarters in which the moving average of the VIX is above (below) the sample median ("VIX") and (b) years in which the CAPM alphas estimated from daily data have a cross-sectional R^2 with pre-1993 CAPM alphas above the median for the post-1993 period ("Alphas"). Arbitrage position is inferred from the abnormal short position on underlying stocks. In the parentheses are t -statistics based on bootstrap standard errors that account for cross-portfolio covariances and generated regressors. Boldface denotes coefficient estimates with the absolute value of t -statistics greater than 1.96.

	Constrained time $\beta_{funding}^{post93}$				Unconstrained time $\beta_{funding}^{post93}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Arbitrage position ^{post93}	2.20 (2.44)		2.43 (2.79)		0.25 (0.20)		0.29 (0.25)	
α_{CAPM}^{pre93}		0.29 (3.00)		0.30 (3.27)		0.03 (0.23)		0.05 (0.38)
$\beta_{funding}^{pre93}$	0.34 (1.56)	−0.14 (−0.53)	0.21 (0.92)	−0.29 (−1.05)	0.20 (0.67)	0.15 (0.40)	0.27 (0.98)	0.19 (0.56)
Constant	−0.12 (−0.09)	−0.35 (−0.25)	−0.25 (−0.20)	−0.52 (−0.39)	−0.15 (−0.09)	−0.18 (−0.10)	−0.02 (−0.01)	−0.04 (−0.02)
Constrained indicator	VIX		Alphas		VIX		Alphas	
Observations	40	40	40	40	40	40	40	40
R_{adj}^2	0.71	0.69	0.79	0.67	0.14	0.12	0.20	0.25

Table 4 strongly favors the arbitrage interpretation over the wealth portfolio interpretation of my previous results. Funding betas during constrained times are large and cross-sectionally explained by both arbitrage position and pre-arbitrage alpha, but they tend to disappear during unconstrained times. Furthermore, although both correlation and volatility can affect beta, my finding is driven by changes in the correlation with the funding factor during constrained times; anomaly return correlations with the funding factor display patterns observed in funding betas (Fig. 5).

3.4. Funding betas as discount rate betas: cross-section of time-series return predictability

Finally, anomaly portfolios with high funding betas feature greater discount rate variations, consistent with the portfolios' funding betas arising from discount rate shocks

that arbitrageurs generate. To show this, I study cross-sectional differences in the time-series predictability of returns. By definition, greater discount rate variation means greater variation in expected returns, so holding all else constant, portfolios with high funding betas should feature greater time-series return predictability. Intuitively, if funding betas arise from the act of arbitrage, high funding beta portfolios should experience greater booms and busts induced by arbitrage capital and exhibit greater return predictability in the time series.²⁷

I test this in a two-stage approach. The first stage is a portfolio-specific time-series predictive regression in which

²⁷ An alternative way to check that a factor beta is a discount rate beta is to decompose stock returns into discount rate vs. cash flow shocks using VAR, as in (Campbell and Vuolteenaho, 2004; Campbell et al., 2009) (CPV). However, as explained in CPV, this approach works most naturally for decomposing yearly returns and is not suitable for my paper, with its relatively short sample period.

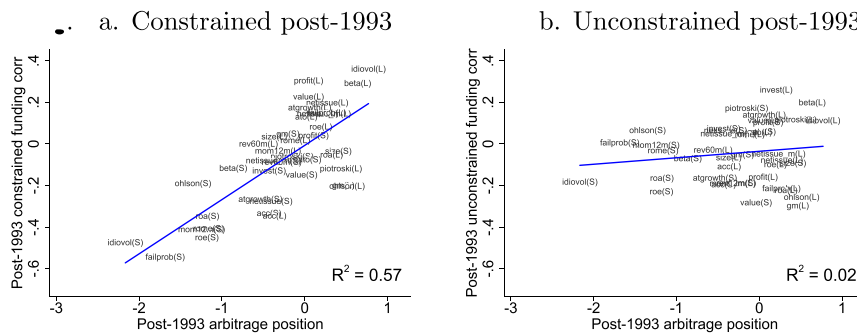


Fig. 5. Funding liquidity exposure arises during constrained times.

The figures show that in the post-1993 period, return correlations with the funding-liquidity factor line up strongly with arbitrage position during constrained periods (left) but not during unconstrained periods (right). The result is similar if I use pre-1993 CAPM alpha as the x-axis. To compute the correlations, I first compute unexplained return as the realized return in excess of the risk-free rate and multivariate (two-factor) market beta times the excess market return. I then take the time-series correlation between the unexplained returns and the funding factor. I use constrained quarters implied by the VIX.

I predict future 12-, 18-, or 24-month cumulative excess returns using its past three or five year cumulative excess return as the predictor (De Bondt and Thaler, 1985; Moskowitz et al., 2012).²⁸

$$r_{i,t \rightarrow t+s}^e = \theta_0 + \theta_1 r_{i,t-L \rightarrow t}^e + \epsilon_{i,t \rightarrow t+s} \quad (s \in \{12m, 18m, 24m\}, L \in \{3y, 5y\}) \quad (10)$$

Past three or five year returns can proxy for valuation ratios such as the book-to-market ratio, often used in return predictability studies, when accounting data are unavailable or are subject to seasonality issues, as is the case for my predictive regressions with monthly data (Fama and French, 1996; Gerakos and Linnainmaa, 2012; Asness et al., 2013). Intuitively, past return predicts future return since high (low) past return means that arbitrageurs have driven up (down) the prices of stocks in the anomaly portfolio at the expense of a lower (higher) expected return going forward. I run my portfolio-specific predictive regression using overlapping monthly data and obtain the R^2 as the measure of how predictable the time series of future portfolio return is.

The second stage is a cross-sectional regression in which I explain the first-stage R^2 using the absolute value of the funding beta, the arbitrage position, or the pre-1993 CAPM alpha of the anomaly portfolio.²⁹ Essentially, this approach allows me to relate funding beta and other measures of arbitrage to the fraction of return volatility coming from discount rate shocks, which the first-stage R^2 captures.

²⁸ The return horizon over which discount-rate changes induced by arbitrage are realized is unclear, so I tried return horizons of one year to three years in the first stage. I find that my second-stage cross-sectional results are robust to using future 30-month return predictive regression and then lose statistical significance from future 36-month return prediction. The past and future returns I use as the left-hand and right-hand variables in the first-stage regression are based on the same stocks that belong to the anomaly portfolio at the time of portfolio formation. Using past and future returns on rebalanced portfolios would be an incorrect approach.

²⁹ I take an absolute value since predictability increases in the magnitude of the discount rate beta, regardless of its sign.

I use my two-stage approach for the post-1993 period, where I expect to find predictability lining up with the absolute value of the funding beta and its determinants, and for the pre-1993 period, which should not feature the same pattern unless return predictability, for whatever reason, is intrinsically correlated with the portfolio's funding beta.

Fig. 6 summarizes my finding. Return predictability increases in the absolute value of the anomaly's funding beta in the post-1993 period but not in the pre-1993 period, consistent with post-1993 funding betas being discount rate betas arising from arbitrage trades. Table 5 shows that this cross-sectional pattern holds with respect to the absolute value of the arbitrage position and pre-1993 alpha as well (Columns (1)–(3)). That is, a portfolio with a larger abnormal return attracts a larger arbitrage position and suffers greater booms and busts due to variation in aggregate arbitrage capital.

The economic magnitude is large. In the baseline specification, an increase of 1 in the absolute value of the funding beta increases the first-stage R^2 of the predictability regression by 0.05–0.06, depending on the return horizon. The R^2 of the second-stage cross-sectional regression reported in the brackets shows that the arbitrage variables explain as much as 58% of the cross-sectional variation in return predictability. The pre-1993 period does not display a cross-sectional relation between predictability and funding beta, suggesting that the large discount rate variation in high funding beta portfolios is unique to the post-1993 period, with its increased arbitrage activity.

These results are consistent with post-1993 funding betas being discount rate exposures—rather than cash flow exposures—to funding liquidity, as the theory predicts. Furthermore, by showing that the time-series return predictabilities of anomaly portfolios are an equilibrium outcome of arbitrage trades, my results add to the growing literature on time-series predictability of anomaly returns. My finding is consistent with (Frazzini and Pedersen, 2014; Huang et al., 2018; Lou and Polk, 2019), who find that the act of arbitrage generates predictable time-series patterns in momentum and low beta stocks.

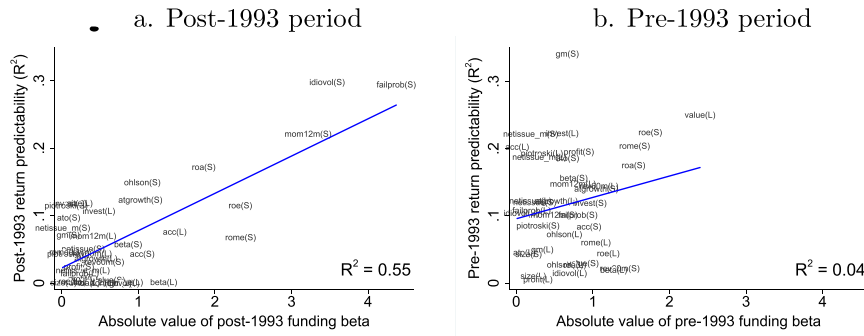


Fig. 6. Cross-section of time-series predictability of returns: post-1993 versus pre-1993.

The figures show that the return predictabilities of anomaly portfolios line up with the absolute value of their funding betas in the post-1993 period (left) but not strongly in the pre-1993 period (right). Return predictability is measured by the R^2 of the portfolio-specific time-series regression that explains future one year cumulative excess returns using the past three year cumulative excess returns.

Table 5

Funding liquidity betas as discount rate betas: evidence from return predictability.

$$1\text{st-stage time-series regression: } r_{i,t \rightarrow t+s}^e = \theta_0 + \theta_1 r_{i,t \rightarrow t}^e + \epsilon_{i,t \rightarrow t+s}, \quad R_{1\text{st stage},i}^2 \equiv \frac{\sum_t (\hat{r}_{i,t \rightarrow t+s}^e - \bar{\hat{r}}_{i,t \rightarrow t+s}^e)^2}{\sum_t (r_{i,t \rightarrow t+s}^e - \bar{r}_{i,t \rightarrow t+s}^e)^2}$$

$$2\text{nd-stage cross-sectional regression (baseline): } R_{1\text{st stage},i}^2 = b_0 + b_1 |\beta_{\text{funding},i}| + u_i$$

This table shows that anomaly portfolios with a larger absolute value of funding beta feature greater return predictability than other portfolios in the post-1993 period, consistent with post-1993 funding betas being discount rate betas arising from the act of arbitrage. Such a pattern does not arise in the pre-1993 period. The regression has two stages. The first stage is a portfolio-specific time-series regression in which I regress the future 12-month, 18-month, or 24-month cumulative excess return $r_{t \rightarrow t+s}^e$ on past 3-year or 5-year cumulative excess return $r_{t-L \rightarrow t}^e$. From the first stage, I obtain the coefficient of determination $R_{1\text{st stage}}^2$ as the measure of time-series predictability of returns. The second stage is a univariate cross-sectional regression in which I regress $R_{1\text{st stage}}^2$ on the absolute value of funding liquidity beta, arbitrage position, or pre-1993 CAPM alpha that predicts the volatility of arbitrage-driven discount rate shocks in the anomaly portfolio. Arbitrage position is inferred from abnormal short position on underlying stocks. Since there are six specifications of the first-stage time-series regression and three right-hand variables in the second-stage cross-sectional regression, each sample period has 18 univariate cross-sectional regressions in total. In the parentheses are t -statistics based on bootstrap standard errors that account for cross-portfolio covariances and generated regressors. In the brackets are the coefficient of determination (R^2) of the second-stage cross-sectional regression. Boldface denotes coefficient estimates with the absolute value of t -statistics greater than 1.96.

1st-stage prediction horizon s:	+12m return		+18m return		+24m return	
1st-stage predictor: Right-hand variable	-3yr return (1)	-5yr return (2)	-3yr return (3)	-5yr return (4)	-3yr return (5)	-5yr return (6)
Panel A: Left-hand variable is the R^2 from 1st-stage predictive regressions in the post-1993 period						
$ \beta_{funding}^{post93} $	0.06 (2.22) [0.55]	0.05 (2.19) [0.55]	0.06 (2.31) [0.52]	0.06 (2.27) [0.54]	0.06 (2.27) [0.48]	0.06 (2.23) [0.51]
$ \text{Arbitrage position}^{post93} $	0.12 (2.33) [0.58]	0.11 (2.27) [0.56]	0.13 (2.60) [0.54]	0.12 (2.48) [0.53]	0.12 (2.47) [0.51]	0.12 (2.37) [0.51]
$ \alpha^{pre93} $	0.02 (2.22) [0.47]	0.02 (2.26) [0.49]	0.02 (2.30) [0.41]	0.02 (2.32) [0.44]	0.02 (2.14) [0.39]	0.02 (2.18) [0.42]
Panel B: Left-hand variable is the R^2 from 1st-stage predictive regressions in the pre-1993 period						
$ \beta_{funding}^{pre93} $	0.03 (0.87) [0.04]	0.04 (1.03) [0.06]	0.04 (1.09) [0.05]	0.04 (1.16) [0.06]	0.05 (1.31) [0.08]	0.05 (1.40) [0.10]
$ \text{Arbitrage position}^{pre93} $	−0.05 (−0.30) [0.01]	−0.04 (−0.21) [0.00]	−0.05 (−0.28) [0.00]	−0.04 (−0.20) [0.00]	−0.01 (−0.08) [0.00]	0.00 (0.02) [0.00]
$ \alpha^{pre93} $	0.00 (0.29) [0.01]	0.00 (0.51) [0.02]	0.00 (0.46) [0.01]	0.00 (0.66) [0.02]	0.01 (0.79) [0.04]	0.01 (1.00) [0.06]

My finding also suggests that the strong time-series predictability of anomaly returns documented in (Haddad et al., 2018) may be unique to the post-1993 sample period with greater arbitrage-driven discount rate shocks to anomalies.

4. Exposure to arbitrageur wealth portfolio shocks

In addition to funding liquidity exposure, the act of arbitrage can also expose assets to shocks coming from other stocks in the arbitrageur wealth portfolio. Indeed, I

show that the anomaly portfolios' betas with respect to my proxy for the arbitrageur wealth portfolio display patterns we expect for arbitrage-driven exposure.

4.1. Graphical evidence

Fig. 2 in Section 1 plots the cross-section of wealth portfolio betas in the pre-1993 and post-1993 periods. In the pre-1993 period with little arbitrage on anomalies, exposures to the arbitrageur wealth portfolio clustered around zero and had no relation to their abnormal return proxied by the CAPM alpha. When arbitrage on anomalies grows in the post-1993 period, however, both arbitrage position and pre-1993 CAPM alpha explain more than 60% of the cross-sectional variation in how much the portfolio return covaries with arbitrageur wealth portfolio shocks. That is, a portfolio with a greater pre-1993 CAPM alpha and greater arbitrage position suffers more when aggregate arbitrageur wealth goes down.

The cross-sectional patterns in the post-1993 period may, however, have an alternative interpretation. These patterns can arise mechanically if a portfolio with a large arbitrage position (i.e., a larger fraction of its market capitalization is held by arbitrageurs) also represents a larger share of the arbitrageur wealth portfolio. Although this interpretation does not explain why these cross-sectional patterns do not arise in the pre-1993 period, a more formal investigation seems helpful.

4.2. Cross-section of wealth portfolio betas

To show more formally that the post-1993 cross-sectional patterns in Fig. 2 do not arise mechanically but rather because of arbitrage, I add the portfolio's market capitalization share in the arbitrageur wealth portfolio ("share of wealth portfolio") as an additional explanatory variable in regressions explaining the cross-section of wealth portfolio betas. If my results were mechanical, the portfolio's share in the arbitrageur wealth should subsume the explanatory power of arbitrage position or pre-1993 CAPM alpha.

Table 6 shows that both arbitrage position and pre-1993 CAPM alpha robustly explain the cross-sectional variation in wealth portfolio betas, regardless of controlling for the portfolio's share in the arbitrageur wealth portfolio. The economic magnitude is large; a 1 pp increase in arbitrage position (in pre-1993 CAPM alpha) is associated with an increase in the wealth portfolio beta by 0.26 to 0.35 (by 0.03 to 0.04).

While the share of arbitrageur wealth portfolio is positively related to wealth portfolio betas, its effect is statistically insignificant. In the pre-1993 period, wealth portfolio betas have little relation to arbitrage position or pre-1993 CAPM alpha, consistent with post-1993 wealth portfolio betas emerging as a consequence of the post-1993 growth in arbitrage capital.

4.3. Additional evidence

Panel A of Table 7 presents similar findings in the panel of wealth portfolio betas. An increase in the arbitrage po-

sition over time increases the anomaly's wealth portfolio betas. Wealth portfolio beta also increases around 1993 in proportion to the portfolio's pre-1993 CAPM alpha but, again, the 1993 dummy and academic publication contribute almost equally to the effect of pre-1993 alpha on wealth portfolio betas such that their relative importance cannot be easily disentangled. The share of wealth portfolio has an estimated coefficient similar to that in the cross-sectional approach but is now statistically significant.

Panel B shows that wealth portfolio betas of anomaly portfolios tend to be larger when arbitrageurs are constrained, although the reduction in betas during unconstrained times depends on the measure I use to proxy for constrained versus unconstrained times. Panel C shows that the time-series return predictabilities in the post-1993 period tend to line up with the absolute value of the wealth portfolio betas, as they do with funding betas, consistent with post-1993 wealth portfolio betas being discount rate betas. These patterns do not hold in the pre-1993 period. Overall, these additional analyses show that arbitrageur wealth portfolio betas also display patterns expected from betas that arise endogenously through arbitrage.

Taken together, the evidence presented in this section shows that a portfolio with a larger abnormal return attracts more arbitrage and attains a larger exposure to arbitrageur wealth portfolio shocks, consistent with the predictions of intermediary-based models.

5. Robustness

Here, I provide several robustness checks to the results in the previous two sections.

5.1. Placebo factors

How likely is it that a spurious factor generates my results on beta exposures of anomaly portfolios? To answer this, I generate 10,000 monthly placebo factors by sorting stocks on a randomly generated number at the end of each month and computing value-weighted returns on a long-short portfolio of stocks that fall into extreme deciles.³⁰ I also generate 10,000 quarterly placebo factors by rolling over monthly returns on monthly placebo factors.

I study the ability of placebo factors to generate the cross-sectional results in Columns (1) and (4) of Tables 2 and 6. That is, I study how well arbitrage position and pre-1993 CAPM alpha explain the cross-section of post-1993 anomaly portfolio betas with respect to a quarterly or monthly placebo factor. For each cross-sectional regression, I compute bootstrap standard errors based on the empirical distribution of 1,000 draws as I do in my original analysis. I repeat this for 10,000 placebo factors to study the distribution of *t*-statistics and *R*² of the regressions.

Panel A of Table 8 shows that it is unlikely for a random factor to generate the strong results I obtain with the funding liquidity and the arbitrageur wealth portfolio factors. The odds of a random quarterly factor generating a

³⁰ I use NYSE stocks to determine the decile cutoffs.

Table 6

Explaining the cross-section of arbitrageur wealth portfolio betas.

Baseline: $\beta_{\text{wealth},i}^{\text{post93}} = b_0 + b_1 \text{Arbitrage position}_i^{\text{post93}} + u_i$

This table shows that the post-1993 arbitrageur wealth portfolio betas of 40 anomaly portfolios can be cross-sectionally explained by arbitrage position or pre-1993 CAPM alpha. Unless otherwise noted, the right-hand variables are calculated for the same sample period as the left-hand variable. Wealth portfolio betas are betas with the arbitrageur wealth portfolio implied by estimated arbitrage positions and are estimated in a two-factor model that includes the market factor. Arbitrage position is inferred from abnormal short position on underlying stocks. Share of wealth portfolio is the portfolio's market capitalization share in the arbitrageur wealth portfolio. Characteristic ranks are the value-weighted decile rank of the underlying stocks' characteristics. Pre-1993 and post-1993 periods are 1974–1993 and 1994–2016, respectively. In the parentheses are *t*-statistics based on bootstrap standard errors that account for cross-portfolio covariances as well as generated regressors. Boldface denotes coefficient estimates with the absolute value of *t*-statistics greater than 1.96.

	$\beta_{\text{wealth}}^{\text{post93}}$								$\beta_{\text{wealth}}^{\text{pre93}}$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Arbitrage position	0.26 (2.73)	0.26 (2.61)	0.35 (2.16)						−0.01 (−0.04)		
$\alpha_{\text{CAPM}}^{\text{pre93}}$				0.03 (3.13)	0.03 (2.80)	0.04 (2.91)				−0.01 (−0.98)	
Share of wealth portfolio		0.46 (0.75)	0.37 (0.52)		0.36 (0.55)	0.48 (0.68)	1.79 (1.84)	1.34 (1.67)	0.33 (0.60)	0.37 (0.72)	0.32 (0.61)
$\beta_{\text{wealth}}^{\text{pre93}}$		0.08 (0.41)			−0.24 (−0.86)		−0.42 (−1.32)				
Size rank			0.04 (0.61)			−0.00 (−0.02)		−0.06 (−0.90)	−0.03 (−0.66)	−0.02 (−1.13)	−0.03 (−1.54)
Value rank			−0.01 (−0.22)			−0.06 (−1.19)		0.03 (0.69)	0.03 (1.02)	0.05 (1.30)	0.03 (1.03)
Profitability rank			−0.01 (−0.61)			−0.04 (−1.12)		0.02 (0.65)	−0.03 (−1.43)	−0.02 (−0.75)	−0.03 (−1.44)
Investment rank			−0.03 (−1.05)			−0.04 (−1.31)		0.00 (0.14)	−0.01 (−0.82)	−0.02 (−1.39)	−0.01 (−1.11)
Constant	−0.00 (−0.07)	0.02 (0.41)	0.15 (0.47)	−0.03 (−0.83)	−0.01 (−0.26)	0.65 (1.46)	0.02 (0.50)	−0.11 (−0.23)	−0.11 (−0.49)	−0.25 (−0.98)	−0.11 (−0.51)
Observations	40	40	40	40	40	40	40	40	40	40	40
R_{adj}^2	0.65	0.66	0.70	0.62	0.64	0.69	0.18	0.19	0.78	0.80	0.78

t-statistic higher than what I obtain from funding liquidity betas (2.31; see Column (1) of Table 2) is 0.87%, slightly lower than 1.05% implied by a *t*-statistic of 2.31.³¹ The adjusted R^2 of 72% is even harder to achieve (0.47%) such that the chance of a random factor outperforming the funding factor in terms of both the *t*-statistic and the adjusted R^2 is 0.27%. My cross-sectional regression explaining post-1993 funding liquidity betas using pre-1993 CAPM alpha (Column (4) of Table 2) is more difficult to replicate with quarterly placebo factors (0.20%).

My monthly cross-sectional regression results are equally difficult to obtain using a random factor. The odds of a random monthly factor generating a *t*-statistic higher than 2.73 in the cross-sectional arbitrage position to beta regression is 0.20%, slightly lower than 0.32% implied by a *t*-statistic of 2.73. The adjusted R^2 of 65% is somewhat easier to achieve (3.59%), but the chance of obtaining this fit while having a *t*-statistic higher than 2.73 is again 0.20%. The cross-sectional alphas-into-betas result (Column (4) of Table 6) is even harder to achieve (0.00%).

To appreciate these results, it is important to remember that my measures of arbitrage capital shocks are suggested by the literature (funding liquidity factor) or by the estimated arbitrage positions (the arbitrageur wealth portfolio) and therefore are not cherry picked. Hence, the joint probability that I fortuitously obtained the strong results

based on funding liquidity betas and arbitrageur wealth portfolio betas must be extremely low.

5.2. The 1993 cutoff and other robustness checks

My findings are robust to alternative choices I could have made in my empirical analysis. My cross-sectional analyses use the year 1993 to proxy for pre- versus post-arbitrage periods for anomalies, but my results are robust to using 1991, 1992, 1994, and 1995 as the end of the pre-arbitrage period. To illustrate, I repeat the main cross-sectional regressions in Tables 2 and 6 using these alternative cutoffs (Panels B and C of Table 8). Across all cutoffs, the coefficients are large and statistically significant. In fact, depending on the regression, I obtain results stronger than those based on the 1993 cutoff.

The Online Appendix shows that my cross-sectional results are also robust to controlling for additional market characteristics of the anomaly portfolios, such as idiosyncratic volatility and market liquidity. This suggests that my results are not a spurious result driven by both the measure of arbitrage activity (arbitrage position or pre-1993 alpha) and post-1993 funding or arbitrage portfolio beta being positively correlated with volatility or market liquidity.

5.3. A test without a factor: evidence from the quant crisis of 2007

Prior tests based on the funding liquidity and arbitrageur wealth portfolio factors require that these factors

³¹ This is the one-sided *p*-value for *t* = 2.31.

Table 7

Additional tests on wealth portfolio betas.

This table repeats the tests in Tables 3, 4, and 5 on betas with respect to the arbitrageur wealth portfolio. In the parentheses are *t*-statistics based on bootstrap standard errors that account for cross-portfolio covariances as well as generated regressors. In Panel B, the numbers in the brackets are the coefficient of determination (R^2) of the second-stage cross-sectional regression. Boldface denotes coefficient estimates with the absolute value off-statistics greater than 1.96.

Panel A: Explaining the panel of arbitrageur wealth portfolio betas											
	OLS								2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Arbitrage position	0.13 (3.58)	0.13 (3.28)							0.58 (2.95)	0.56 (2.51)	0.62 (2.73)
$\alpha_{CAPM}^{pre93} \times \text{Post-1993}$			0.03 (2.47)	0.03 (2.50)			0.02 (1.82)				
$\alpha_{CAPM}^{pre93} \times \text{Post publication}$					0.04 (2.66)	0.04 (2.71)	0.02 (1.81)				
Share of wealth portfolio		0.15 (1.81)		0.37 (3.46)		0.37 (3.41)	0.36 (3.30)	0.44 (3.65)	-0.87 (-2.05)	-0.82 (-1.75)	-0.96 (-1.98)
Anomaly FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control for characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	17,280	17,280	17,280	17,280	17,280	17,280	17,280	17,280	17,280	17,280	17,280
R^2_{adj}	0.15	0.17	0.18	0.21	0.18	0.20	0.23	0.09	.	.	.
Instrumental variables											
$\alpha_{CAPM}^{pre} \times \text{Post-1993}$									✓	✓	
$\alpha_{CAPM}^{pre} \times \text{Postpub}$									✓		✓

Panel B: Wealth portfolio betas tend to arise during constrained times								
	Constrained time β_{wealth}^{post93}				Unconstrained time β_{wealth}^{post93}			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Arbitrage position	0.24 (2.07)		0.35 (2.59)		0.25 (2.26)		0.06 (0.62)	
α_{CAPM}^{pre93}		0.03 (2.33)		0.04 (2.82)		0.03 (2.50)	0.01 (0.54)	
Share of wealth portfolio	0.73 (0.57)	0.79 (0.63)	0.62 (0.40)	0.05 (0.03)	1.21 (0.94)	0.69 (0.45)	1.57 (0.97)	1.74 (1.09)
β_{wealth}^{pre93}	0.16 (0.65)	-0.15 (-0.48)	-0.08 (-0.27)	-0.50 (-1.38)	0.01 (0.06)	-0.29 (-0.95)	0.33 (1.26)	0.24 (0.84)
Constant	0.01 (0.37)	-0.00 (-0.10)	0.02 (0.39)	-0.02 (-0.39)	-0.00 (-0.00)	-0.04 (-0.64)	0.01 (0.26)	0.01 (0.17)
Constrained indicator	VIX		Alphas		VIX		Alphas	
Observations	40	40	40	40	40	40	40	40
R^2_{adj}	0.58	0.58	0.63	0.62	0.63	0.59	0.22	0.20

Panel C: Wealth portfolio betas as discount rate betas: evidence from return predictability						
LHS: R^2 from 1st-stage predictive regressions, RHS: $ \beta_{wealth} $						
1st-stage predicted variable:	+12m return		+18m return		+24m return	
1st-stage predictor variable:	-3yr return	-5yr return	-3yr return	-5yr return	-3yr return	-5yr return
Sample period	(1)	(2)	(3)	(4)	(5)	(6)
Post-1993	0.32 (2.10) [0.52]	0.32 (2.08) [0.51]	0.34 (2.25) [0.47]	0.33 (2.22) [0.49]	0.32 (2.11) [0.43]	0.32 (2.10) [0.46]
Pre-1993	-0.12 (-0.77) [0.01]	-0.09 (-0.61) [0.01]	0.03 (0.22) [0.00]	0.04 (0.24) [0.00]	0.17 (1.11) [0.02]	0.17 (1.10) [0.02]

correctly represent shocks to the capital of anomaly arbitrageurs. Here, I provide further evidence on the cross-sectional relation between arbitrage position (or pre-1993 alpha) and exposure to arbitrage capital shocks without taking a stand on the arbitrage capital factor. The idea is to focus on a single severe arbitrage capital shock. Since the cross-sectional variation in returns during such an event would be mostly driven by differences in beta exposure

to arbitrage capital, I can test if the cross-section of returns during the event line up with the theoretical determinants of arbitrage-driven betas. To do this, I use the crash of quantitative long/short equity hedge fund capital in August 2007, which I describe briefly before proceeding to my tests.

Over the three-day period of August 7–9, 2007, seemingly distinct equity portfolios commonly underperformed.

Table 8

Robustness.

Panel A reports the likelihood that betas with respect to a randomly generated factor portfolio exhibit the patterns I find for funding liquidity and wealth portfolio betas. I use 10,000 randomly generated placebo factors to calculate the likelihood. Panels B and C repeat the test in Columns (1) and (4) of Tables 2 and 6 using alternative cutoff years in the early 1990s. In the parentheses are *t*-statistics based on bootstrap standard errors that account for cross-portfolio covariances as well as generated regressors. Boldface denotes coefficient estimates with the absolute value of *t*-statistics greater than 1.96.

Panel A: What is probability that a random placebo factor generates my results?								
	<i>t</i> -statistic		<i>R</i> ²		<i>t</i> -stat & <i>R</i> ² Jointly			
Arbitrage position to funding liquidity betas:	0.87%		0.47%		0.27%			
Pre-93 CAPM alpha to funding liquidity betas:	0.47%		0.60%		0.20%			
Arbitrage position to wealth portfolio betas:	0.20%		3.59%		0.20%			
Pre-93 CAPM alpha to wealth portfolio betas:	0.05%		0.75%		0.00%			

Panel B: Using alternatives cutoffs to explain post-cutoff funding liquidity betas								
	1991		1992		1994		1995	
Arbitrage position ^{post}	1.38		1.36		1.91		1.80	
	(2.24)		(2.19)		(2.62)		(2.60)	
α_{CAPM}^{pre}		0.16		0.17		0.24		0.24
		(2.39)		(2.49)		(2.75)		(2.72)
$\beta_{funding}^{pre}$	0.34	0.14	0.34	0.13	0.27	−0.18	0.28	−0.17
	(1.71)	(0.52)	(1.82)	(0.44)	(0.99)	(−0.44)	(1.03)	(−0.44)
Constant	−0.15	−0.30	−0.15	−0.29	−0.19	−0.40	−0.18	−0.35
	(−0.85)	(−1.16)	(−0.77)	(−1.07)	(−0.82)	(−1.15)	(−0.79)	(−1.06)
Observations	40	40	40	40	40	40	40	40
<i>R</i> _{adj} ²	0.77	0.73	0.74	0.71	0.79	0.66	0.78	0.69

Panel C: Using alternatives cutoffs to explain post-cutoff arbitrageur wealth portfolio betas								
	1991		1992		1994		1995	
Arbitrage position ^{post}	0.28		0.27		0.26		0.26	
	(2.76)		(2.67)		(2.57)		(2.60)	
α_{CAPM}^{pre}		0.03		0.03		0.03		0.03
		(2.95)		(2.83)		(2.72)		(2.73)
Share of wealth portfolio	0.41	0.37	0.40	0.34	0.44	0.37	0.40	0.36
	(0.71)	(0.57)	(0.66)	(0.52)	(0.70)	(0.56)	(0.65)	(0.59)
β_{wealth}^{pre}	0.06	−0.17	0.10	−0.17	0.09	−0.22	0.10	−0.22
	(0.30)	(−0.69)	(0.50)	(−0.65)	(0.43)	(−0.77)	(0.46)	(−0.78)
Constant	0.01	−0.02	0.01	−0.01	0.01	−0.01	0.01	−0.01
	(0.23)	(−0.38)	(0.27)	(−0.32)	(0.37)	(−0.25)	(0.33)	(−0.17)
Observations	40	40	40	40	40	40	40	40
<i>R</i> _{adj} ²	0.68	0.69	0.66	0.65	0.66	0.62	0.67	0.64

Fig. 7a shows that anomaly portfolios that arbitrageurs typically go long on commonly posted losses (in 17 out of 20 cases) and portfolios that arbitrageurs typically go short on commonly posted gains (in 19 out of 20 cases). Hedge funds that take long-short positions on these anomalies also suffered severe losses. Fig. 7b shows that these arbitrageurs suffered a cumulative loss of almost 6% over August 7–9, and then the return rebounded over the following three days (August 10–14). Remarkably, the crash and the recovery were exclusive to equity anomalies; other arbitrage strategies remained unaffected.

The primary explanation for this unusual covariance event in the anomalies is a systematic drop in arbitrage capital. It is speculated that following a portfolio underperformance since early July 2007, one or more arbitrageurs rapidly unwound their arbitrage position on anomalies, possibly due to margin calls (Khandani and Lo, 2007; Khandani and Lo, 2011; Pedersen, 2012; Stein, 2009). This led to losses by other arbitrageurs that in turn triggered more margin calls until anomaly arbitrageurs commonly suffered capital losses.

Treating the three-day crash period of the crisis as the period in which the level of arbitrage capital dropped severely, I ask whether returns on different portfolios during the crash can be explained cross-sectionally by the differences in their arbitrage position or pre-arbitrage alpha, analogous to my analysis on betas. Furthermore, since this drop in the asset price is a discount rate (valuation) shock rather than cash flow shock, I also ask whether the return during the three-day recovery period can also be explained cross-sectionally.

Fig. 8 summarizes my findings. Cumulative returns on anomaly portfolios during the crisis are cross-sectionally and strongly explained by their post-1993 arbitrage position and pre-1993 CAPM alpha. This is consistent with the key mechanism that generates a cross-section of arbitrage-driven betas: an asset with a more positive (negative) pre-arbitrage alpha, and hence positive (negative) arbitrage position, drops (gains) more in response to a sharp decline in arbitrage capital. An opposite pattern holds during recovery, consistent with anomaly portfolio

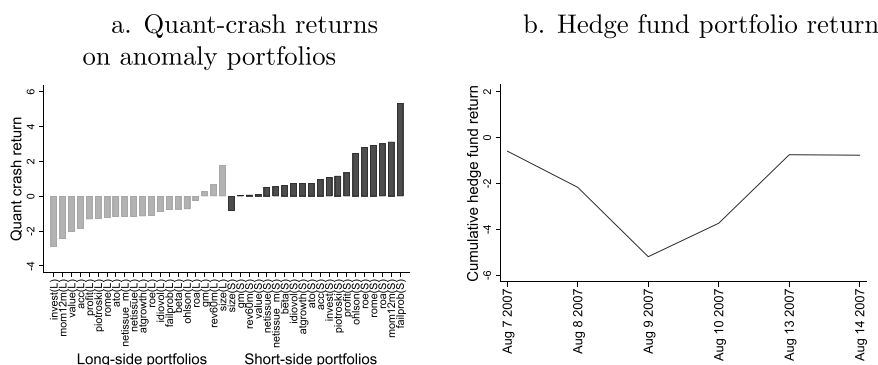


Fig. 7. Description of the quant crisis of August 2007.

The first figure plots the cumulative three day returns of long-side (first 20) and short-side (next 20) anomaly portfolios during the quant crash of August 7–9, 2007. The second figure plots the cumulative daily returns on equity market-neutral hedge funds during the entire quant crisis period, which includes both the crash (August 7–9) and recovery (August 10–14) periods. The hedge fund return data are from Hedge Fund Research.

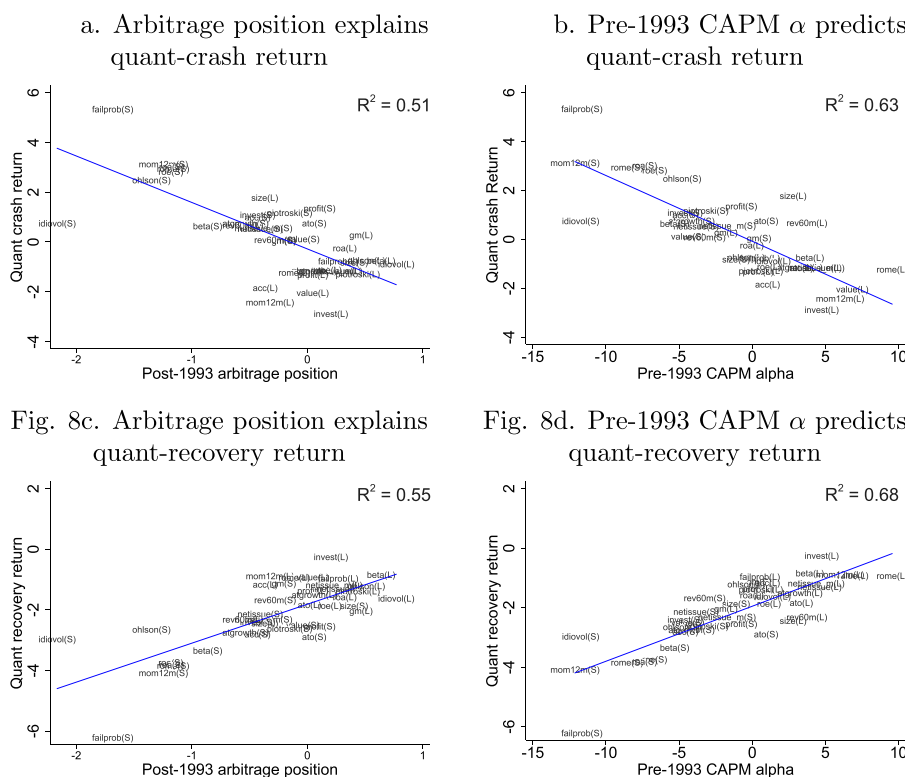


Fig. 8. Explaining the cross-section of returns during quant crash and recovery.

The figures show that post-1993 arbitrage position and pre-1993 CAPM alpha explain the cross-section of anomaly portfolio returns during the quant crash (August 7–9, 2007; top two figures) and recovery (August 10–14; bottom two figures).

liers' quant crisis returns being discount rate movements.³² Table B.2 shows that this result is robust to inferring arbitrage position from the month before the crash (July 2007) and to using cumulative abnormal returns instead of raw returns.

³² The cross-sectional pattern of long anomalies earning positive returns and short anomalies earning negative returns during the recovery is clearer in abnormal returns than in raw returns.

5.4. Closing remarks

I close with three additional points about my results. First, the interpretation of my results does not depend heavily on whether equity anomalies actually represent hidden risk, mispricing, or measurement error. What matters is that institutional arbitrageurs, such as hedge funds, trade anomalies, regardless of the debate. In fact, even if anomaly returns represent rational compensation for risk,

arbitrage-driven betas would arise through risk sharing (Kondor and Vayanos, 2019). Intuitively, anomalies with larger arbitrageur positions rely more heavily on the risk-sharing role of arbitrageurs and hence become more sensitive to arbitrage capital. Also, even if some anomalies were measurement errors, a high past alpha that occurs by chance can still attract arbitrage capital and give rise to an arbitrage-driven beta. In this case, however, the arbitrage-driven beta would eventually disappear after arbitrageurs realized that the anomaly was a measurement error and stopped trading it.

Second, one may suggest that once arbitrage has driven down the anomaly alphas, arbitrage-driven betas should no longer arise. This is not the case. If the original source of the demand distortion remains, alphas can remain low only in the presence of arbitrage trades that generate the arbitrage-driven betas. In other words, in the presence of systematic shocks to arbitrage capital, low alphas and nonzero arbitrage-driven betas should coexist in equilibrium.

Third, my results do not necessarily imply that intermediaries who act as arbitrageurs make financial assets riskier. For example, it could be that equity anomalies arise because of undiversifiable risks that households face but arbitrageurs do not and when arbitrageurs enter the market, they reduce the equilibrium risk of assets by providing risk-sharing. In this case, the act of arbitrage reduces the risk of the assets for households and increases their risk for the arbitrageurs until risk premia associated with these two measures of risk are equalized. However, my results do imply that once arbitrageurs turn alphas into betas, fundamental mispricing defined as the deviation of price from fundamental value can persist in the form of arbitrage-driven risk rather than abnormal returns.

6. Conclusion

This paper shows that financial intermediaries that act as arbitrageurs in the asset market play a crucial role in determining the equilibrium risk of assets, consistent with the implications of asset pricing models that emphasize the role of intermediary arbitrageurs. I show this in the context of equity anomaly portfolios, using funding liquidity and arbitrageur wealth portfolio shocks to measure risk from the perspective of arbitrageurs. My results suggest that asset pricing anomalies are, at least in part, alphas turned into betas.

My findings in the equity market suggest that arbitrageurs may play similar risk determination roles in other asset classes. Understanding the sources of equilibrium risk exposure to arbitrage capital shocks in other asset markets would nicely complement the growing evidence that these risk exposures are priced in the cross-section of assets and provide a more complete account of the role of financial intermediaries in the asset market.

Appendix A. Bootstrap procedure

To understand my bootstrap procedure, consider a statistic,

$$\hat{b}(\mathbf{X}; G) = \hat{b}(X_1, \dots, X_N, F; G)$$

$$= \hat{b}\left(\left(\begin{array}{c} x_{11} \\ \vdots \\ x_{1T} \end{array}\right), \dots, \left(\begin{array}{c} x_{N1} \\ \vdots \\ x_{NT} \end{array}\right), \left(\begin{array}{c} f_1 \\ \vdots \\ f_T \end{array}\right); G\right),$$

where $\mathbf{X}$ denotes the full panel data used in my analysis; $X_i = (x_{i1} \dots x_{iT})'$ denotes a time series of observations for portfolio i and time periods $t = 1, \dots, T$; $x_{it} = (r_{it}, \text{arb position}_{it}, \dots)$ is a column of observations pertaining to portfolio i at time t ; and F is the time-series realizations of aggregate factors. G denotes parameters that govern the joint distribution of the sample of data $\mathbf{X} = (X_1, \dots, X_N, F)$. It contains information about the cross-sectional covariances among $(X_1, \dots, X_N, F)$ and about the time-series variation within X_i or F . To provide more context, $\hat{b}$ could be the Ordinary Least Squares estimator for the slope coefficient on arbitrage position (1) of Table 2):

$$\beta_{\text{funding},i}^{\text{post93}} = b_0 + b_1 \text{Arbitrage position}_i^{\text{post93}} + u_i.$$

In the population, $\text{Var}(\hat{b})$ is driven by both cross-sectional and time-series variations in the draw of $(X_1, \dots, X_N, F)$. That is, when each sample $\mathbf{X}_s = (X_1^s, \dots, X_N^s, F^s)$ is drawn from the population, (a) a set of N portfolios is chosen from a population of anomaly portfolios and (b) time-series data for the chosen portfolios and factors are drawn from a population of time-series data. This way, each sample $\mathbf{X}_s$ preserves the distributional characteristics summarized by G . And for each sample $\mathbf{X}_s$, we would compute the value of the statistic $\hat{b}^s \equiv \hat{b}(\mathbf{X}_s)$. $\text{Var}(\hat{b})$ is then given by the variance of $\hat{b}^1, \dots, \hat{b}^\infty$ corresponding to an infinite draw of samples.

My nonparametric bootstrap approach simulates the procedure described in the previous paragraph but uses an empirical distribution G^* rather than the true population distribution G (Efron, 1979). To do this, I construct each bootstrap sample $\mathbf{X}_b^*$ by (a) randomly drawing a new set of 40 portfolios from the existing set of 40 anomaly portfolios with replacement and (b) randomly drawing time-series observations (with replacement) from existing time-series data on anomaly portfolios and factors. This way, each bootstrap sample $\mathbf{X}_b^*$ preserves the distributional characteristics G that generated the original sample, and the collection of bootstrap samples $G^* = \{\mathbf{X}_1^*, \dots, \mathbf{X}_{1,000}^*\}$ serves as the empirical distribution from which $\text{Var}(\hat{b})$ is estimated. For each bootstrap sample $\mathbf{X}_b^*$, I compute $\hat{b}(\mathbf{X}_b^*)$. My bootstrap estimator for $\text{Var}(\hat{b})$, denoted $\hat{\text{Var}}(\hat{b})^{\text{bootstrap}}$, is then estimated as the variance of $\hat{b}(\mathbf{X}_1^*), \dots, \hat{b}(\mathbf{X}_{1,000}^*)$.

To summarize, my bootstrap procedure has four steps:

1. For each bootstrap b , randomly draw with replacement the set of anomaly portfolios I_b^* : e.g.,

$$I = (1, 2, 3, 4, 5, \dots, 40) \rightarrow I_b^* = (17, 4, 1, 1, 33, \dots, 28)$$

2. Also randomly draw with replacement the time-series data for portfolios and factors while maintaining the cross-sectional dependence by drawing the entire row

from each time period: e.g.,

$$\mathbf{X} = \left(\begin{pmatrix} x_{1,1} \\ \vdots \\ x_{1,33} \\ \vdots \\ x_{1,T} \end{pmatrix}, \begin{pmatrix} x_{2,1} \\ \vdots \\ x_{2,33} \\ \vdots \\ x_{2,T} \end{pmatrix}, \dots, \begin{pmatrix} x_{N,1} \\ \vdots \\ x_{N,33} \\ \vdots \\ x_{N,T} \end{pmatrix}, \begin{pmatrix} f_1 \\ \vdots \\ f_{33} \\ \vdots \\ f_T \end{pmatrix} \right)$$

$$\rightarrow \mathbf{X}_b^* = \left(\begin{pmatrix} x_{17,33} \\ x_{17,9} \\ \vdots \\ \vdots \\ \vdots \end{pmatrix}, \begin{pmatrix} x_{4,33} \\ x_{4,9} \\ \vdots \\ \vdots \\ \vdots \end{pmatrix}, \dots, \begin{pmatrix} x_{28,33} \\ x_{28,9} \\ \vdots \\ \vdots \\ \vdots \end{pmatrix}, \begin{pmatrix} f_{33} \\ f_9 \\ \vdots \\ \vdots \\ \vdots \end{pmatrix} \right)$$

The time-series data are constructed by drawing the first 80 rows (240 rows for monthly data) from the pre-1993 period and the next 92 rows (276 rows for monthly data) from the post-1993 period.

- Repeat steps 1 and 2 to construct 1,000 bootstrap samples $\mathbf{X}_1^*, \dots, \mathbf{X}_{1,000}^*$. These samples represent the empirical distribution $G^* = \{\mathbf{X}_1^*, \dots, \mathbf{X}_{1,000}^*\}$.
- Bootstrap variance of the estimator $\hat{b}$ is given by the empirical distribution of the estimator obtained from each bootstrap sample:

$$\widehat{Var}(\hat{b})^{bootstrap} = Var(\hat{b}(\mathbf{X}_1^*), \dots, \hat{b}(\mathbf{X}_{1,000}^*))$$

The square root of the bootstrap variance is the bootstrap standard error.

For regressions involving both quarterly and monthly data, I first obtain the random draw of quarters to construct quarterly bootstrap data and then construct monthly bootstrap data based on monthly observations pertaining to the quarters used in the quarterly bootstrap. For panel regressions, I draw moving blocks of four quarters with replacement from each subsample of ten years to retain time-series dependencies in the variables (Knsch, 1989) and to account for the possible publication effect in addition to the post-1993 effect.³³ For quant crisis regressions, I cannot generate bootstrap samples using in-sample data since the quant crisis is a single observation. Instead, I use two months surrounding the quant crisis (July 6, 2007 to September 14, 2007) to draw three day returns with replacement. Alternative approaches such as using previous one month returns or previous one year returns generate stronger or similarly strong results.

In addition to standard errors, the bootstrap also provides an estimate of the downward bias in the coefficient due to generated regressors. Adjusting for these biases tend to increase the estimated coefficients and t -statistics by around 10%, but to be conservative, I report parameter estimates and t -statistics that do not adjust for the bias.

Appendix B. Additional tables

³³ Since my sample has 43 years, the last subsample of 2004–2016 has 13 years. However, the exact way I apply the moving block bootstrap does not materially affect the standard errors in my panel regressions.

Table B.1

Determinants of arbitrage position.

Baseline: Arbitrage position_{*i,t*} = $b_0 + b_1\alpha_i^{pre} \times \mathbf{1}(t > 1993q4) + b_2\mathbf{1}(t > 1993q4) + b_3t + b_4t^2 + u_i + \epsilon_{i,t}$. This table shows that pre-1993 alpha predicts post-1993 and post-publication arbitrage position in a panel regression (40 portfolios $\times$ 1974q1–2016q4). The dependent variable measures arbitrage position on portfolio i in quarter t inferred from abnormal short positions on underlying stocks (Section 2.2). The post-1993 dummy is zero for the pre-1993 period (1974q1–1993q4) and one for the post-1993 period (1994q1–2016q4). A portfolio's pre-arbitrage alpha, denoted α^{pre} , is measured by its pre-1993 alpha with respect to the factor model specified in the column heads. For failure probability, α^{pre} is computed from 1981 onward to account for the portfolio's sensitivity to sample period, emphasized in Dichev (1998). Post publication, Post sample, Post-1993, and Post-1993 $\times$ Post-publication (whenever appropriate) as well as quadratic time trends (t and t^2) and a constant are included in the regression but are not reported in the table. In the parentheses are t -statistics based on bootstrap standard errors that account for cross-portfolio covariances, generated regressors, and serial correlations. Boldface denotes coefficient estimates with the absolute value of t -statistics greater than 1.96.

	$\alpha^{pre} = \text{CAPM alpha}$					$\alpha^{pre} = \text{FF3 alpha}$			$\alpha^{pre} = \text{FF5 alpha}$			Long vs. short			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
$\alpha^{pre} \times \text{Post-1993}$	0.076 (3.92)		0.058 (3.12)	0.051 (1.97)	0.052 (2.52)	0.058 (3.15)	0.055 (2.14)	0.045 (2.18)	0.066 (3.56)	0.061 (2.36)	0.051 (2.47)				
$\alpha^{pre} \times \text{Post publication}$		0.076 (3.61)	0.032 (1.78)	0.019 (0.98)	0.027 (1.11)	0.034 (1.89)	0.029 (1.52)	0.012 (0.49)	0.044 (2.47)	0.036 (1.88)	0.022 (0.90)				
$\alpha^{pre} \times \text{Post sample}$				0.025 (0.94)			0.010 (0.36)			0.016 (0.60)					
$\alpha^{pre} \times \text{Post-1993} \times \text{Postpub}$					0.010 (0.27)			0.034 (0.96)			0.036 (1.00)				
Long $\times$ Post-1993												0.643 (3.71)		0.534 (2.97)	0.507 (2.21)
Long $\times$ Post publication													0.597 (3.28)	0.195 (1.16)	0.152 (0.92)
Long $\times$ Post sample															0.082 (0.35)
Anomaly FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,880	6,880	6,880	6,880	6,880	6,880	6,880	6,880	6,880	6,880	6,880	6,880	6,880	6,880	6,880
R^2_{adj}	0.20	0.16	0.21	0.21	0.21	0.27	0.27	0.28	0.27	0.27	0.28	0.16	0.13	0.16	0.17

Table B.2

Explaining the cross-section of quant-crisis returns.

$$\text{Baseline: } r_i^{\text{quant crash}} = b_0 + b_1 \text{Arbitrage position}_i^{\text{post93}} + u_i$$

This table shows that the exposure of anomaly portfolios to arbitrage capital shocks measured by cumulative returns during the quant crash (August 7–9, 2007) is cross-sectionally explained by arbitrage position and pre-1993 CAPM alpha. Cumulative returns during the recovery from the crash (August 10–14, 2007) display an opposite pattern, suggesting that the portfolio returns during the crash were discount rate shocks. Cumulative abnormal return is defined as the excess return net of market exposure (market excess return times the beta estimated over the two months surrounding the quant crisis, July 7–August 6 and August 15–September 14, using three day returns). Post-1993 arbitrage position is computed over the entire post-1993 period. In the parentheses are *t*-statistics based on bootstrap standard errors that account for cross-portfolio covariances and generated regressors. Boldface denotes coefficient estimates with the absolute value of *t*-statistics greater than 1.96.

	Quant-crash return			Quant-crash abnormal return			Quant-recovery return			Quant-recovery abnormal return		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Arbitrage position ^{post93}	−1.87 (−3.32)			−1.99 (−3.69)			1.29 (2.13)			0.94 (1.62)		
Arbitrage position ^{July2007}		−1.28 (−3.03)			−1.39 (−3.30)			0.92 (2.03)			0.62 (1.37)	
$\alpha^{\text{pre93}}_{\text{CAPM}}$			−0.27 (−4.33)			−0.28 (−4.87)			0.18 (2.90)			0.16 (2.62)
Constant	−0.28 (−1.15)	−0.27 (−1.02)	−0.08 (−0.28)	0.34 (1.37)	0.34 (1.21)	0.56 (1.92)	−1.82 (−1.00)	−1.81 (−0.99)	−1.95 (−1.06)	0.01 (0.00)	−0.01 (−0.00)	−0.07 (−0.04)
Observations	40	40	40	40	40	40	40	40	40	40	40	40
R_{adj}^2	0.50	0.35	0.62	0.54	0.40	0.64	0.54	0.41	0.67	0.34	0.22	0.58

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